Aurélien Nicosia

Academic curriculum vitae

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Chargé d'enseignement (permanent teaching appointment), Department of Mathematics and Statistics

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Profile

I develop computational statistical methods for model diagnostics and uncertainty quantification, with a focus on latent-state models, directional data and density estimation. My work connects statistical methodology, simulation, visualization and reproducible R software. My experience combines research, university teaching in French and English, academic coordination and data science leadership in industry.

Education

- 2024. PhD in Statistics, Université Laval. Dissertation: *General Multi-State Models for the Analysis of Animal Movement*. Supervisors: Thierry Duchesne and Louis-Paul Rivest. Dissertation.
- 2010-2013. MSc in Probability and Statistics, Université Toulouse III. Thesis on random matrices and the smallest singular value. Master's thesis, Toulouse 2013, corrected edition 2026 (PDF).
- 2007-2010. BSc in Mathematics, Université Toulouse III.

Academic and professional experience

- Since 2022. Université Laval, Department of Mathematics and Statistics: *chargé d'enseignement* (permanent teaching appointment) and coordinator of the Mathematics and Statistics Help and Learning Centre (CDA).
- Since 2021. Université du Québec à Chicoutimi: lecturer in statistics, data science and programming; supervision of undergraduate internships and research projects.
- 2020-2022. Co-operators: Director, Research and Analytics Innovation; led data science projects and created the Innovation Space in partnership with UQAC.
- 2017-2020. GSK Canada: Data Science Manager; process modelling, data governance and traceability in a regulated pharmaceutical environment.
- 2014-2017. Université Laval: statistical consultant for research teams across several faculties.

Research themes

- Sequential models: hidden Markov models, step-selection functions, generative validation and uncertainty in latent states.
- Directional statistics: angular regression, random-effects models and visualization of dependence.
- Density diagnostics: leave-one-out influence, outlier detection and grid-based computation.

Research program and reproducible materials.

Peer-reviewed articles

- Nicosia, A. (2026). Extra hidden states under emission misspecification in hidden Markov models. Statistica Neerlandica, 80(4), e70042. https://doi.org/10.1111/stan.70042
- Amicizia, D., Marchini, F., Schenone, I., Massaro, E., Landa, P., Nicosia, A., Orsi, A., Icardi, G., Ansaldi, F. (2026). Adjuvanted Influenza Vaccines Versus Other Influenza Vaccines in Adults Aged ≥60 Years: A Systematic Review and Meta-Analysis of Relative Effectiveness for Preventing Influenza-Related Hospitalization. Vaccines, 14(10), 883. https://doi.org/10.3390/vaccines14100883
- Nicosia, A. (2026). Visualizing directional dependence on cylinders and tori. Journal of Data Science, Statistics, and Visualisation, 6(7). https://doi.org/10.52933/jdssv.v6i7.181
- Nicosia, A. (2026). Beyond the next step: A multi-criteria generative validation framework for step selection functions. Methods in Ecology and Evolution, 17(6), 1754-1767. https://doi.org/10.1111/2041-210X.70313
- Gagnon, S., Allard, M., Nicosia, A. (2018). Diurnal and seasonal variations of tundra CO_2 emissions in a polygonal peatland near Salluit, Nunavik, Canada. Arctic Science, 4(1), 1-15. https://doi.org/10.1139/AS-2016-0045
- Nicosia, A., Duchesne, T., Rivest, L.-P., Fortin, D. (2017). A Multi-State Conditional Logistic Regression Model for the Analysis of Animal Movement. Annals of Applied Statistics, 11(3), 1537-1560. https://doi.org/10.1214/17-AOAS1045
- Nicosia, A., Duchesne, T., Rivest, L.-P., Fortin, D. (2017). A General Hidden State Random Walk Model for Animal Movement. Computational Statistics & Data Analysis, 105, 76-95. https://doi.org/10.1016/j.csda.2016.07.009
- Rivest, L.-P., Duchesne, T., Nicosia, A., Fortin, D. (2016). A General Angular Regression Model for the Analysis of Data on Animal Movement in Ecology. Journal of the Royal Statistical Society Series C: Applied Statistics, 65(3), 445-463. https://doi.org/10.1111/rssc.12124

Published discussion in statistics education

- Nicosia, A. (2026). Discussion of "Addressing the Challenges of AI-Generated Assignment Submissions in Education: Insights and Strategies". Journal of Data Science, 24(1), 276-281. https://doi.org/10.6339/26-JDS1208H

Preprints

Manuscripts released on arXiv or bioRxiv, listed separately from published journal articles.

- Nicosia, A., Duchesne, T., Carbon, M. (2026). A Leave-One-Out Influence Statistic for Density-Based Outlier Detection. arXiv:2607.14335. doi:10.48550/arXiv.2607.14335.
- Nicosia, A. (2026). Identifiability and likelihood estimation for homogeneous angular regression with random coefficients. arXiv:2607.06790. doi:10.48550/arXiv.2607.06790.
- Nicosia, A. (2026). Tropical Viterbi Tubes for Decoding Uncertainty in Hidden Markov Models. arXiv:2606.17181. doi:10.48550/arXiv.2606.17181.
- Nicosia, A. (2026). Sequential predictive e-diagnostics for hidden Markov models of animal movement. bioRxiv. doi:10.64898/2026.07.07.737005.
- Bouderbala, I., Nicosia, A., Fortin, D. (2026). Behavioural state inference from movement and environmental data using Markovian step selection functions. bioRxiv. doi:10.64898/2026.02.05.704063.

Software and open resources

Packages released on CRAN

- CircularRegression: circular regression.
- ggcircular: circular, axial and directional visualization with ggplot2.
- GLBFP: density estimation with linear-blend frequency polygons. Leave-one-out diagnostics are in the GitHub development version.
- tutorizeR: interactive tutorial creation from R Markdown or Quarto.
- DonutMap: donut-chart mapping.
- GeneralOaxaca: generalized Blinder-Oaxaca decomposition.
- UlavalSSD: Quebec data and tools for introductory data science; version 0.3.1, published October 2, 2026.

Projects released on GitHub

- learnrTrackR: tracking and export of interactive tutorial attempts.
- leaflet.indoor: experimental multi-level indoor mapping package.
- lmmix: Gaussian mixed models with correlated residuals, under development.
- Données bleues: Quebec and Canadian datasets with reproducible teaching activities.

Selected presentations

- 2026. *A Leave-One-Out Influence Statistic for Scalable Density-Based Outlier Detection*, with Thierry Duchesne and Michel Carbon, Statistical Society of Canada annual meeting. Presentation.
- 2026. Workshop on AI and teaching, Statistical Society of Canada annual meeting. Resource gallery.
- September 18, 2026. *Au-delà du prompt*, Department of Operations and Decision Systems seminar, Université Laval. Presentation and resources, in French.

Teaching and academic coordination

- Université Laval: STT-1100, Introduction to Data Science; STT-2200, Data Analysis; STT-4230/STT-6230, Programming for Data Science; STT-1900, Statistics for Engineering; STT-2920, Probability for Engineering; MQT-2101, Data Analysis and Modelling.
- GSO-6092, *Project Management*, graduate level: taught in English in Winter 2026 and Fall 2026 (in progress).
- GSO-6082, Gestion de projets, graduate level: partial replacement assignment of 21 teaching hours, August 31 to October 16, 2026 (in progress).
- UQAC: 8GEN444, Engineering Statistics; 8STT108, Statistical Analysis of Large Data Sets; 8INF404, Introduction to Data Science and Business Intelligence; 8INF416, Data Visualization and Interface Design.
- STT-1900: in-person and online teaching and coordination for approximately 350 students in 2025.
- CDA: coordination and training of approximately 30 teaching assistants; development of diagnostic assessments in mathematics, numerical analysis and probability to guide student support.

Supervision and examination

- Faculty consultant for three undergraduate projects: directional statistics (2024), random matrices (2025) and visual inference (2026).
- Supervision of industry internships and undergraduate research projects at UQAC.
- Examiner for two master's theses.

Teaching funding and awards

- 2025 and 2026. GPT-CDA and GPT-CDA, Phase 2: two PAIP teaching innovation grants, artificial intelligence and teaching stream, CAD 20,000 each, Université Laval.
- 2026. New Investigator Presentation Award, Statistical Society of Canada, for the presentation on leave-one-out influence for outlier detection.
- 2022-2023 through 2025-2026. Teaching Stars, Faculty of Science and Engineering, Université Laval.
- 2026. Student Recognition, Teaching Award, Gala du mérite.
- 2016. Presentation award, Statistical Society of Canada; Student Symposium winner, SAS Global Forum.
- 2014. Professor Star Award, Université Laval.

Academic service and program development

- Contributed to the review of the statistics degree and the development of the bachelor's program in statistics and data science at Université Laval.
- Produced and graded the 20-MB-A2 Probability and Statistics examination for the Ordre des ingénieurs du Québec.
- Invited in 2025 to a Canadian Engineering Accreditation Board meeting on common courses and student support for 14 engineering programs.
- Co-adaptation of *Probabilités et statistique, Science et génie*, fourth edition, forthcoming from TC Média Livres in April 2027.

Languages

University teaching, scientific writing and communication in French and English.